



Phase 1: Domain Analysis Foundation

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Outline

- Objective & Pipeline
- Phase 1 Methodology
- Experimental Results
- Next Steps

Objective

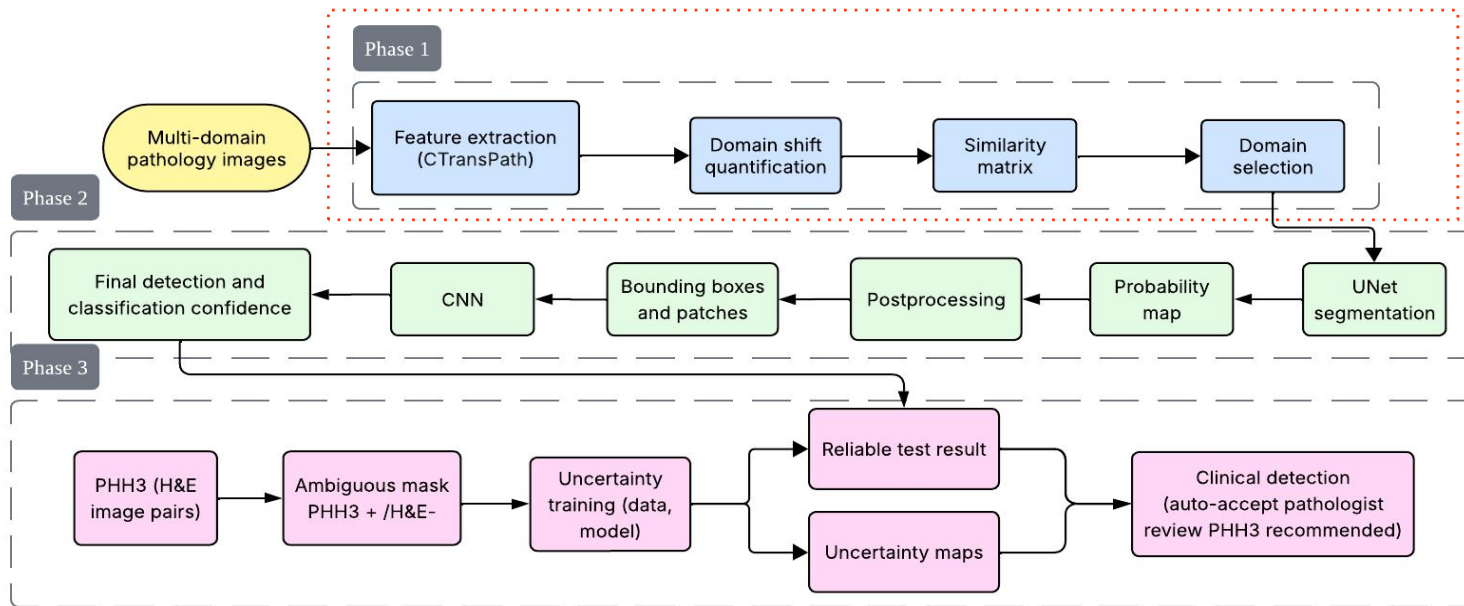
To develop a systematic pipeline to analyze domain relationships, select optimal training domains, and build a robust, ambiguity-aware detection model.

Phase 1: Quantitative Domain Analysis & Strategic Selection

Phase 2: U-Net segmentation & Classification Pipeline

Phase 3: PHH3 Ambiguity Modeling & Uncertainty Quantification

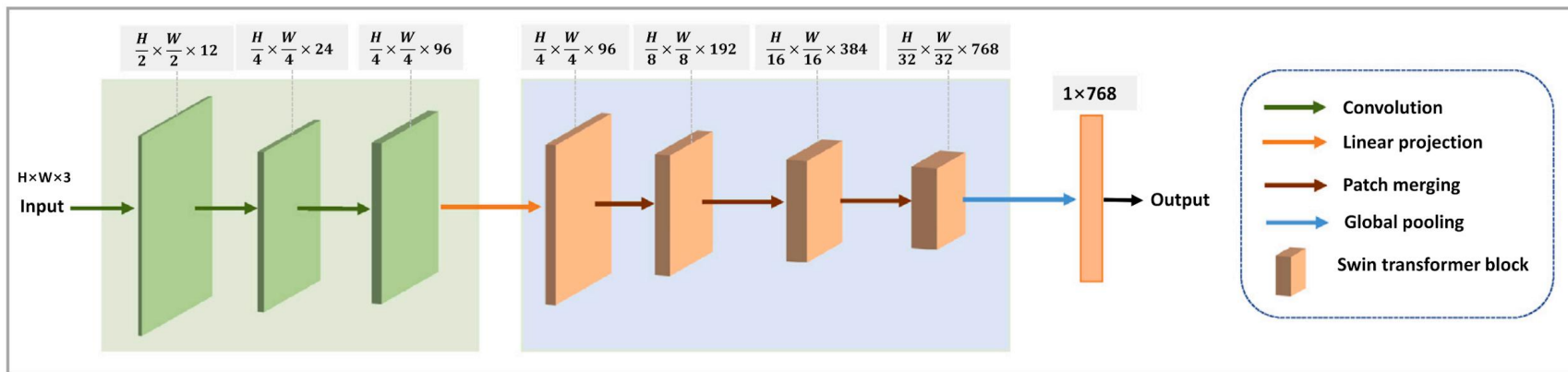
Pipeline



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Feature Extraction: CTransPath



- Proposing a novel self-supervised learning strategy, Semantically-Relevant Contrastive Learning (SRCL), to learn from unlabeled histopathological images.
- Utilizing CTransPath, a hybrid backbone integrating a CNN and a Swin Transformer, to capture both local fine structures and global context.
- Achieving state-of-the-art, transferable performance on five downstream tasks, including patch classification, WSI classification, **mitosis detection**.

Patch Extractions from WSI

- Goal: This patch-based approach is used instead of resizing the entire WSI. Resizing would destroy critical cellular-level details and lose the fine-grained information needed for the model.

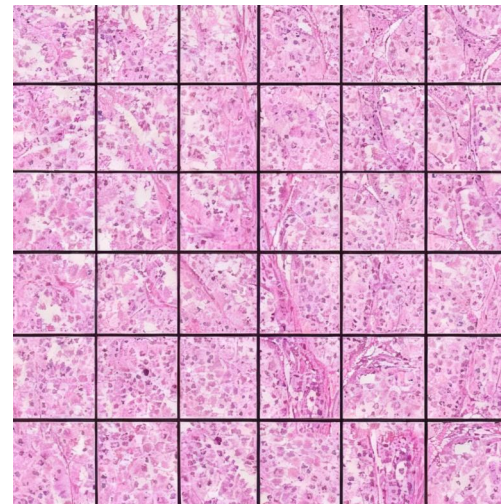
- Methodology :

- Patch Size: 224×224 pixels.

- Stride: 224 pixels (resulting in non-overlapping tiles).

- Filter: Non-white proportion > 0.1

- More than 10% tissue



Dimensionality Reduction

Goal:

To visualize the high-dimensional feature space from CTransPath in 2D. This allows us to visually inspect domain relationships and identify natural clusters.

How?

We use UMAP (Uniform Manifold Approximation and Projection).

- Input: The feature vectors from CTransPath, standardized using StandardScaler.
- Output: A 2-component (2D) embedding for each patch.

Domain shift quantification

Maximum Mean Discrepancy (MMD) - Quantifies the difference between two probability distributions; goal to minimize between training and test datasets

Correlation Alignment (CORAL) - Aligns second-order statistics of source and target distributions, effective when no labeled data is available

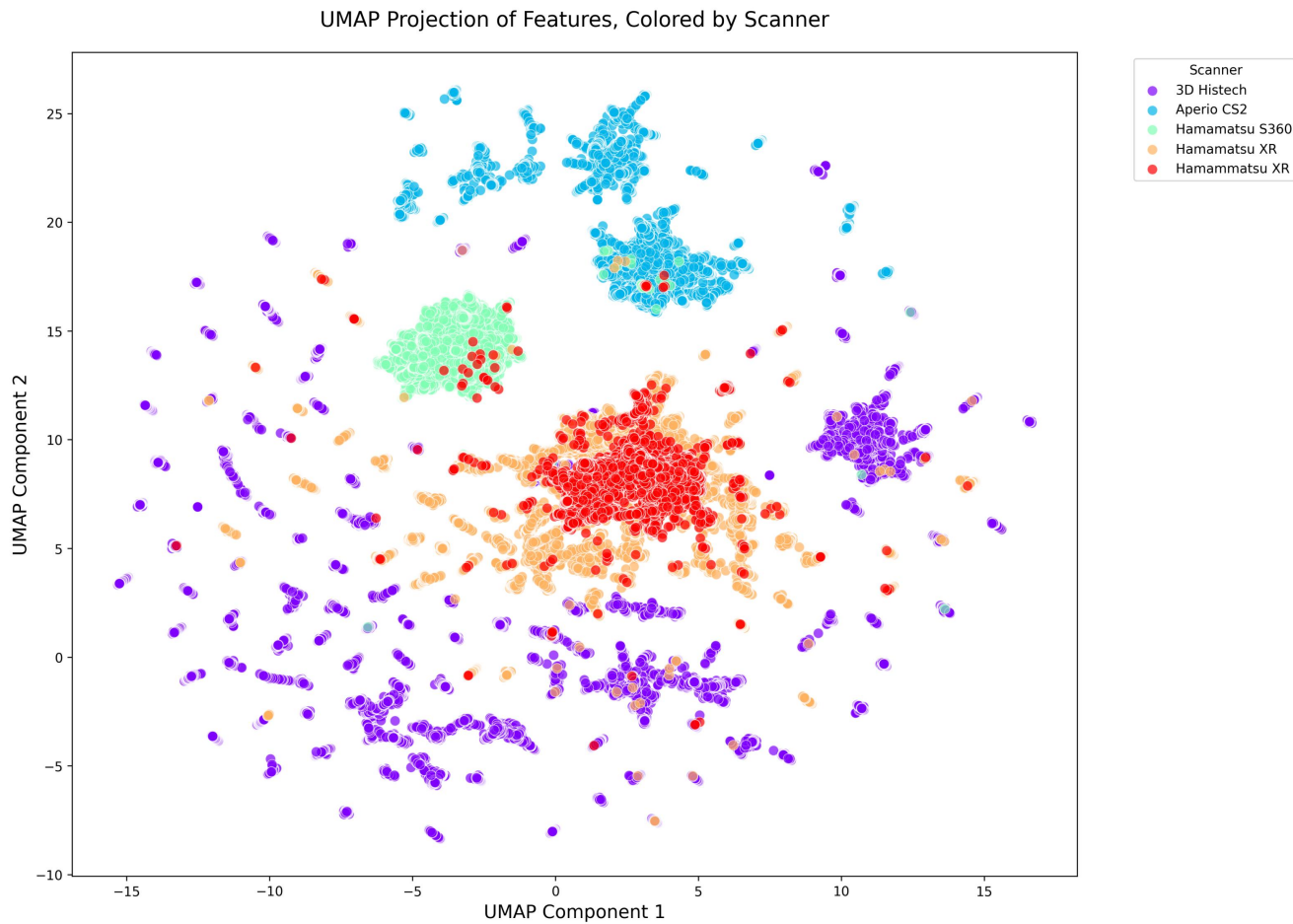
Proxy A-distance - Uses proxy variables to identify causal effects in settings where latent variables are not readily observed

Wasserstein Distance - Distance function defined between probability distributions; quantifies the cost of transforming one distribution into another by calculating optimal transport plan

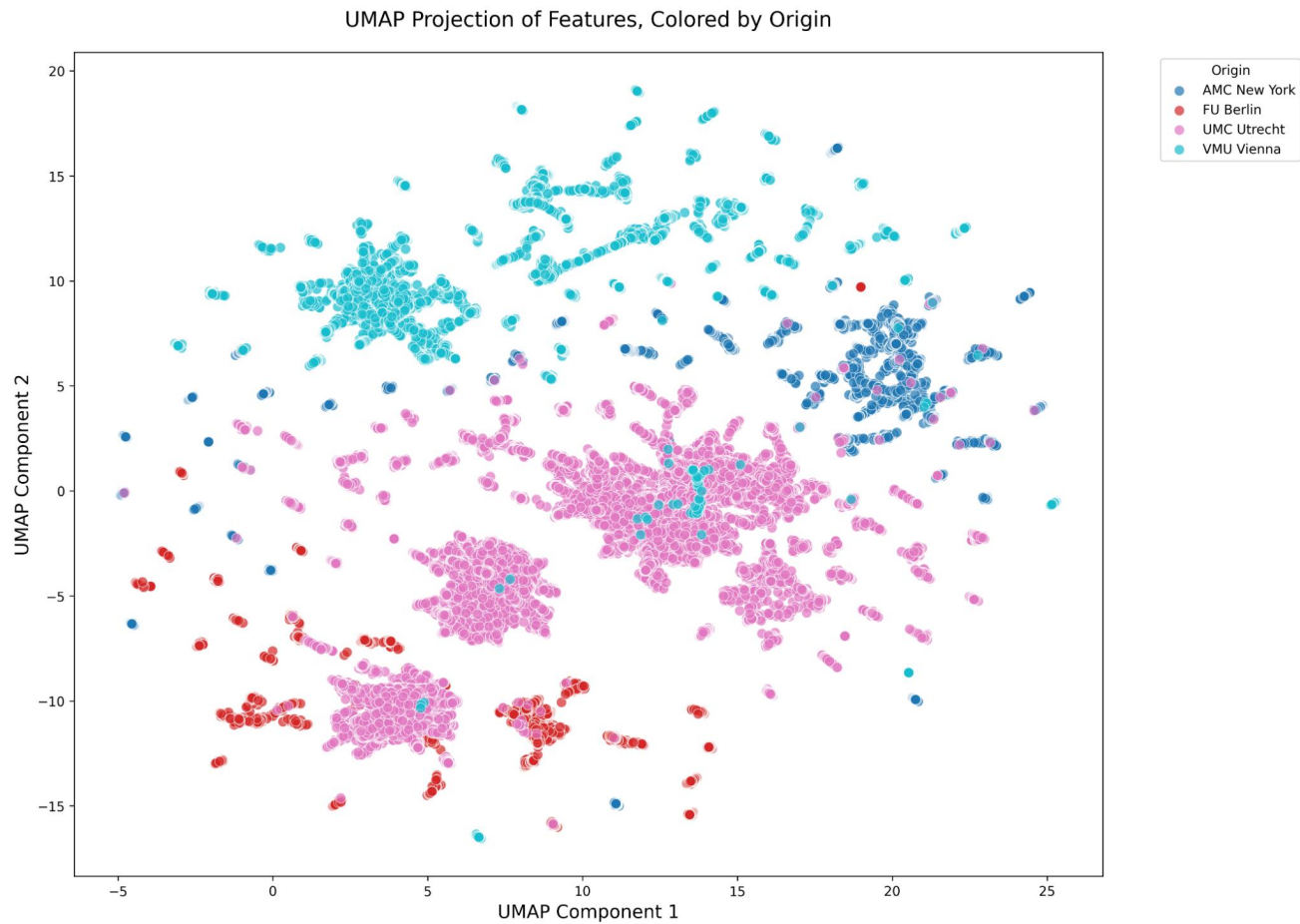
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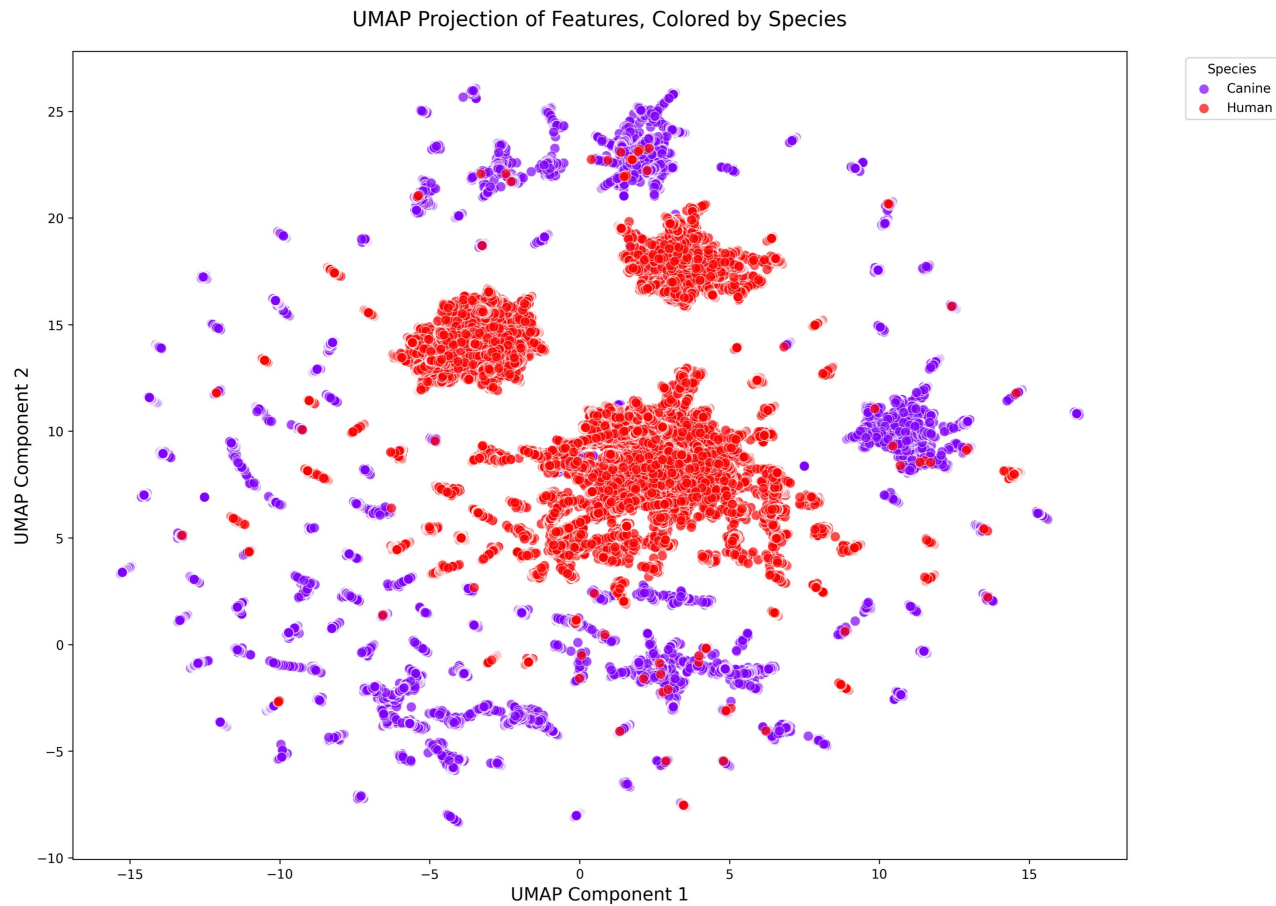
UMAP: Scanner



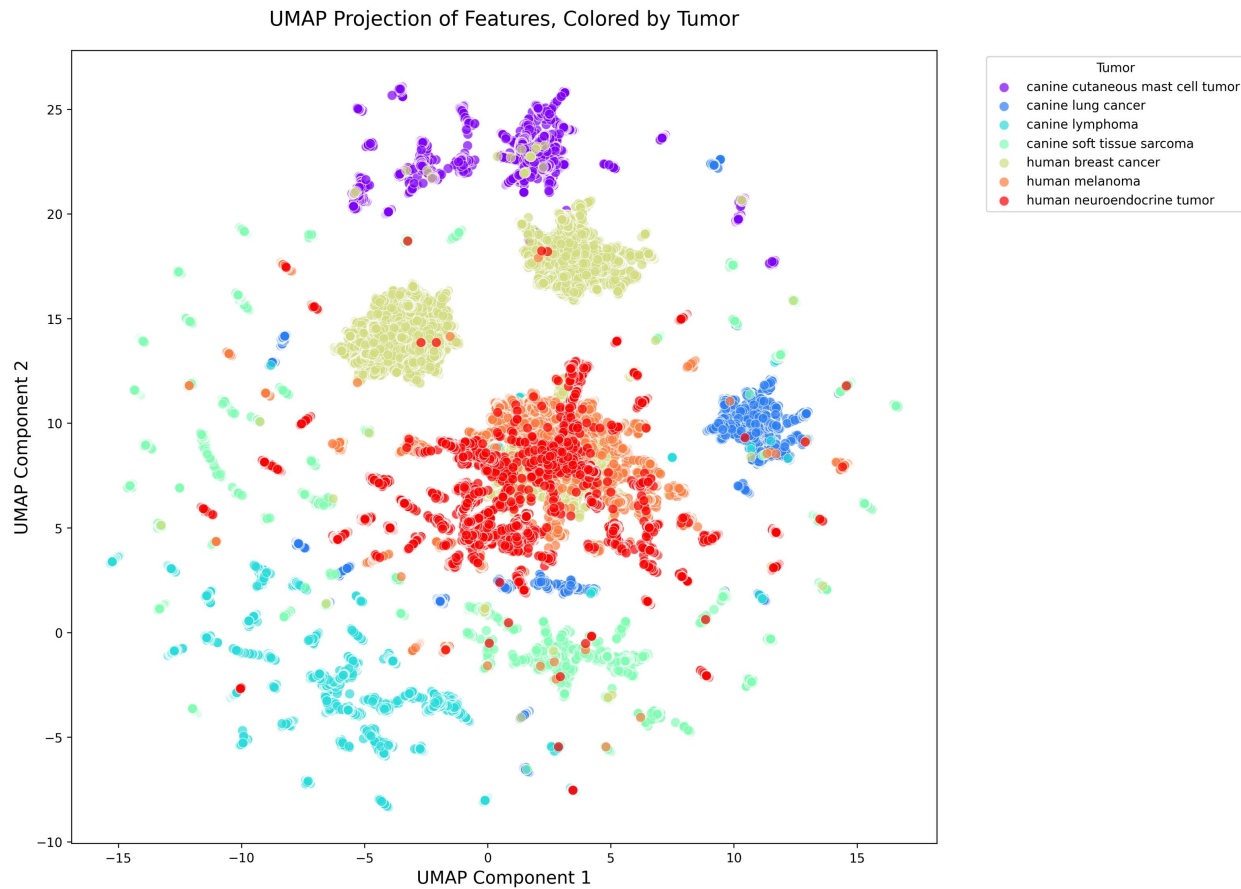
UMAP: Lab Origin



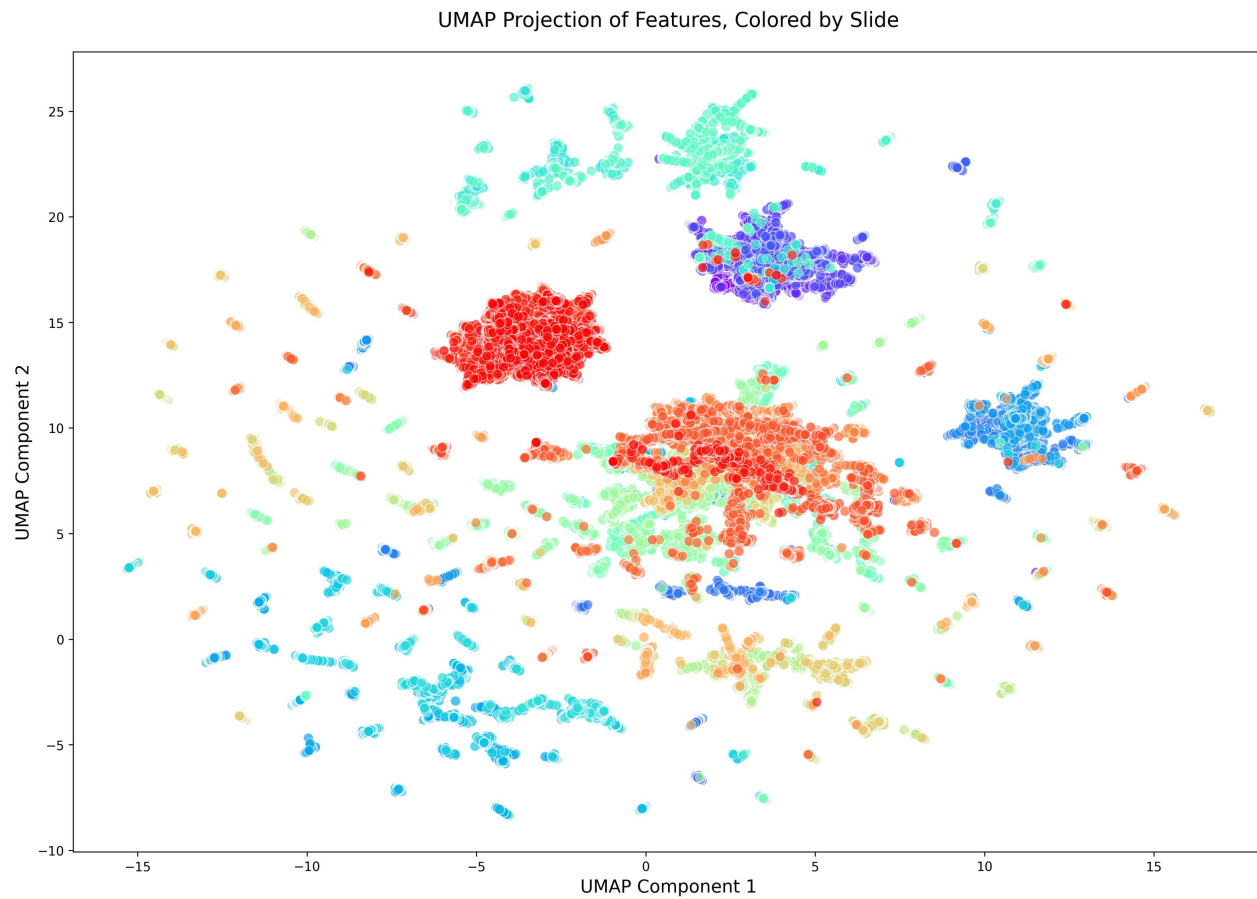
UMAP: Species



UMAP: Tumor Type



UMAP: WSI



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Next Steps

Step 1:

- Normalize the slides to flatten curve
- redo the CTransPath and UMAP to see if clusters are closer together
 - see if scanner and lab origin clusters reduce in distance
 - better visualize morphological features

Step 2:

- Feature extraction on unbiased model (EfficientNet)
- Repeat with normalized slides

Step 3:

- Determine if there are natural clusters within the feature space (Potential areas to start training data selection)
- Play around with UMAP parameter to see if there are better visualizations

Step 4:

- Determine an algorithm for weights on each measurement parameter

Thank you!